

1.2 Convergence

We make a transition from the arithmetic and geometric properties of complex numbers described above to the key notions of convergence and limits.

A sequence $\{z_1, z_2, \dots\}$ of complex numbers is said to **converge** to $w \in \mathbb{C}$ if

$$\lim_{n \rightarrow \infty} |z_n - w| = 0, \quad \text{and we write} \quad w = \lim_{n \rightarrow \infty} z_n.$$

This notion of convergence is not new. Indeed, since absolute values in $\mathbb{C}$ and Euclidean distances in $\mathbb{R}^2$ coincide, we see that z_n converges to w if and only if the corresponding sequence of points in the complex plane converges to the point that corresponds to w .

As an exercise, the reader can check that the sequence $\{z_n\}$ converges to w if and only if the sequence of real and imaginary parts of z_n converge to the real and imaginary parts of w , respectively.

Since it is sometimes not possible to readily identify the limit of a sequence (for example, $\lim_{N \rightarrow \infty} \sum_{n=1}^N 1/n^3$), it is convenient to have a condition on the sequence itself which is equivalent to its convergence. A sequence $\{z_n\}$ is said to be a **Cauchy sequence** (or simply **Cauchy**) if

$$|z_n - z_m| \rightarrow 0 \quad \text{as } n, m \rightarrow \infty.$$

In other words, given $\epsilon > 0$ there exists an integer $N > 0$ so that $|z_n - z_m| < \epsilon$ whenever $n, m > N$. An important fact of real analysis is that $\mathbb{R}$ is complete: every Cauchy sequence of real numbers converges to a real number.¹ Since the sequence $\{z_n\}$ is Cauchy if and only if the sequences of real and imaginary parts of z_n are, we conclude that every Cauchy sequence in $\mathbb{C}$ has a limit in $\mathbb{C}$. We have thus the following result.

Theorem 1.1 *$\mathbb{C}$, the complex numbers, is complete.*

We now turn our attention to some simple topological considerations that are necessary in our study of functions. Here again, the reader will note that no new notions are introduced, but rather previous notions are now presented in terms of a new vocabulary.

1.3 Sets in the complex plane

If $z_0 \in \mathbb{C}$ and $r > 0$, we define the **open disc** $D_r(z_0)$ of radius r centered at z_0 to be the set of all complex numbers that are at absolute

¹This is sometimes called the Bolzano-Weierstrass theorem.

value strictly less than r from z_0 . In other words,

$$D_r(z_0) = \{z \in \mathbb{C} : |z - z_0| < r\},$$

and this is precisely the usual disc in the plane of radius r centered at z_0 . The **closed disc** $\overline{D}_r(z_0)$ of radius r centered at z_0 is defined by

$$\overline{D}_r(z_0) = \{z \in \mathbb{C} : |z - z_0| \leq r\},$$

and the boundary of either the open or closed disc is the circle

$$C_r(z_0) = \{z \in \mathbb{C} : |z - z_0| = r\}.$$

Since the **unit disc** (that is, the open disc centered at the origin and of radius 1) plays an important role in later chapters, we will often denote it by $\mathbb{D}$,

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}.$$

Given a set $\Omega \subset \mathbb{C}$, a point z_0 is an **interior point** of Ω if there exists $r > 0$ such that

$$D_r(z_0) \subset \Omega.$$

The **interior** of Ω consists of all its interior points. Finally, a set Ω is **open** if every point in that set is an interior point of Ω . This definition coincides precisely with the definition of an open set in $\mathbb{R}^2$.

A set Ω is **closed** if its complement $\Omega^c = \mathbb{C} - \Omega$ is open. This property can be reformulated in terms of limit points. A point $z \in \mathbb{C}$ is said to be a **limit point** of the set Ω if there exists a sequence of points $z_n \in \Omega$ such that $z_n \neq z$ and $\lim_{n \rightarrow \infty} z_n = z$. The reader can now check that a set is closed if and only if it contains all its limit points. The **closure** of any set Ω is the union of Ω and its limit points, and is often denoted by $\overline{\Omega}$.

Finally, the **boundary** of a set Ω is equal to its closure minus its interior, and is often denoted by $\partial\Omega$.

A set Ω is **bounded** if there exists $M > 0$ such that $|z| < M$ whenever $z \in \Omega$. In other words, the set Ω is contained in some large disc. If Ω is bounded, we define its **diameter** by

$$\text{diam}(\Omega) = \sup_{z, w \in \Omega} |z - w|.$$

A set Ω is said to be **compact** if it is closed and bounded. Arguing as in the case of real variables, one can prove the following.

Theorem 1.2 *The set $\Omega \subset \mathbb{C}$ is compact if and only if every sequence $\{z_n\} \subset \Omega$ has a subsequence that converges to a point in Ω .*

An **open covering** of Ω is a family of open sets $\{U_\alpha\}$ (not necessarily countable) such that

$$\Omega \subset \bigcup_{\alpha} U_{\alpha}.$$

In analogy with the situation in $\mathbb{R}$, we have the following equivalent formulation of compactness.

Theorem 1.3 *A set Ω is compact if and only if every open covering of Ω has a finite subcovering.*

Another interesting property of compactness is that of **nested sets**. We shall in fact use this result at the very beginning of our study of complex function theory, more precisely in the proof of Goursat's theorem in Chapter 2.

Proposition 1.4 *If $\Omega_1 \supset \Omega_2 \supset \cdots \supset \Omega_n \supset \cdots$ is a sequence of non-empty compact sets in $\mathbb{C}$ with the property that*

$$\text{diam}(\Omega_n) \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

then there exists a unique point $w \in \mathbb{C}$ such that $w \in \Omega_n$ for all n .

Proof. Choose a point z_n in each Ω_n . The condition $\text{diam}(\Omega_n) \rightarrow 0$ says precisely that $\{z_n\}$ is a Cauchy sequence, therefore this sequence converges to a limit that we call w . Since each set Ω_n is compact we must have $w \in \Omega_n$ for all n . Finally, w is the unique point satisfying this property, for otherwise, if w' satisfied the same property with $w' \neq w$ we would have $|w - w'| > 0$ and the condition $\text{diam}(\Omega_n) \rightarrow 0$ would be violated.

The last notion we need is that of connectedness. An open set $\Omega \subset \mathbb{C}$ is said to be **connected** if it is not possible to find two disjoint non-empty open sets Ω_1 and Ω_2 such that

$$\Omega = \Omega_1 \cup \Omega_2.$$

A connected open set in $\mathbb{C}$ will be called a **region**. Similarly, a closed set F is connected if one cannot write $F = F_1 \cup F_2$ where F_1 and F_2 are disjoint non-empty closed sets.

There is an equivalent definition of connectedness for open sets in terms of curves, which is often useful in practice: an open set Ω is connected if and only if any two points in Ω can be joined by a curve γ entirely contained in Ω . See Exercise 5 for more details.

2 Functions on the complex plane

2.1 Continuous functions

Let f be a function defined on a set Ω of complex numbers. We say that f is **continuous** at the point $z_0 \in \Omega$ if for every $\epsilon > 0$ there exists $\delta > 0$ such that whenever $z \in \Omega$ and $|z - z_0| < \delta$ then $|f(z) - f(z_0)| < \epsilon$. An equivalent definition is that for every sequence $\{z_1, z_2, \dots\} \subset \Omega$ such that $\lim z_n = z_0$, then $\lim f(z_n) = f(z_0)$.

The function f is said to be continuous on Ω if it is continuous at every point of Ω . Sums and products of continuous functions are also continuous.

Since the notions of convergence for complex numbers and points in $\mathbb{R}^2$ are the same, the function f of the complex argument $z = x + iy$ is continuous if and only if it is continuous viewed as a function of the two real variables x and y .

By the triangle inequality, it is immediate that if f is continuous, then the real-valued function defined by $z \mapsto |f(z)|$ is continuous. We say that f attains a **maximum** at the point $z_0 \in \Omega$ if

$$|f(z)| \leq |f(z_0)| \quad \text{for all } z \in \Omega,$$

with the inequality reversed for the definition of a **minimum**.

Theorem 2.1 *A continuous function on a compact set Ω is bounded and attains a maximum and minimum on Ω .*

This is of course analogous to the situation of functions of a real variable, and we shall not repeat the simple proof here.

2.2 Holomorphic functions

We now present a notion that is central to complex analysis, and in distinction to our previous discussion we introduce a definition that is genuinely *complex* in nature.

Let Ω be an open set in $\mathbb{C}$ and f a complex-valued function on Ω . The function f is **holomorphic at the point** $z_0 \in \Omega$ if the quotient

$$(1) \quad \frac{f(z_0 + h) - f(z_0)}{h}$$

converges to a limit when $h \rightarrow 0$. Here $h \in \mathbb{C}$ and $h \neq 0$ with $z_0 + h \in \Omega$, so that the quotient is well defined. The limit of the quotient, when it exists, is denoted by $f'(z_0)$, and is called the **derivative of f at z_0** :

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}.$$